# Standard Operating Procedure (SOP)
# General Repair Technician: MicroBT WhatsMiners

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## 1. Purpose

This Standard Operating Procedure (SOP) establishes standardized guidelines and best practices for MicroBT Whatsminer diagnosis, repair, and maintenance. This SOP aims to ensure consistent, high-quality repairs that maximize equipment uptime, extend hardware lifespan, and maintain optimal mining performance while adhering to safety standards and manufacturer specifications.

## 2. Scope

This SOP applies to all repair technicians responsible for diagnosing, troubleshooting, and repairing MicroBT Whatsminer equipment, including but not limited to M-series models (M20, M21, M30, M31, M50 series, etc.). The procedures cover component-level repairs, board replacements, firmware updates, and comprehensive performance testing across all repair operations.

# 3. Responsibilities

## 3.1 Repair Technician

- Diagnose hardware and firmware issues in Whatsminer equipment
- Perform repairs according to established procedures
- Document all repair actions, parts used, and test results
- Maintain a clean and organized workspace
- Follow all safety protocols and procedures
- Report recurring issues or improvement opportunities
- Maintain and properly use all required tools and equipment
- Complete required quality control testing before returning equipment to service

## 3.2 Repair Department Manager

- Oversee repair operations and ensure compliance with SOPs
- Manage repair technician schedules and workloads
- Review repair documentation and quality control metrics
- Approve non-standard repair procedures when necessary
- Ensure adequate inventory of repair parts and supplies
- Coordinate with other departments regarding repair status and timelines
- Implement continuous improvement initiatives for repair processes

## 3.3 Quality Control Specialist

- Conduct final testing of repaired equipment
- Verify repair documentation completeness
- Maintain repair quality standards
- Track repair success rates and recurring issues
- Provide feedback to technicians on repair quality

## 3.4 Inventory Manager

- Maintain adequate stock of replacement parts
- Track parts usage and forecast needs
- Process RMA (Return Merchandise Authorization) for defective parts
- Verify authenticity of replacement components
- Manage disposal of non-repairable components

# 4. Definitions

- **ASIC: Application-Specific Integrated Circuit;** specialized chip designed specifically for cryptocurrency mining
- **Hash Board:** Circuit board containing multiple ASIC chips that perform the actual mining calculations
- **Control Board:** Main circuit board that controls the operation of the miner
- **PSU:** Power Supply Unit that provides electrical power to the miner
- **Hashrate:** Computational power measurement, expressed in hashes per second (H/s, KH/s, MH/s, GH/s, TH/s, PH/s)
- **Firmware:** Software programmed into hardware devices that provides control instructions
- **WhatsMiner Tool:** Software utility used to configure and monitor Whatsminer equipment
- **MicroBT Interface:** Built-in web management interface for configuring and monitoring Whatsminer devices
- **DPS:** Damage Protection System, hardware/firmware feature in newer models
- **RMA:** Return Merchandise Authorization for returning defective parts to manufacturer
- **ESD:** Electrostatic Discharge; potentially damaging electrical discharge to sensitive components
- **Microsoldering:** Precision soldering techniques used for small components on circuit boards
- **Hot-air Rework:** Technique using controlled hot air for removing and replacing surface-mounted components
- **PIC:** Programmable Interface Controller used in some Whatsminer control boards

# 5. Required Qualifications and Skills

## 5.1 Educational Background

- Technical certification or associate degree in electronics, computer hardware, or related field
- MicroBT certification (if available) or equivalent training
- Basic electrical engineering knowledge

## 5.2 Experience

- Minimum 1 year experience in electronics repair, preferably with mining equipment
- Demonstrated experience in board-level diagnostics and repair
- Experience with soldering and microsoldering techniques

## 5.3 Technical Skills

- Proficiency in reading and interpreting circuit schematics
- Advanced soldering skills (through-hole and surface mount components)
- Ability to use diagnostic tools and equipment

- Understanding of networking basics and IP configuration
- Familiarity with MicroBT firmware and management interfaces
- Basic programming and scripting knowledge (optional but beneficial)

### 5.4 Soft Skills

- Methodical problem-solving approach
- Strong attention to detail
- Ability to document work thoroughly and accurately
- Time management skills
- Communication skills for status reporting and team collaboration
- Ability to work independently and in a team environment

## 6. Equipment and Tools

### 6.1 Personal Protective Equipment (PPE)

- ESD wrist strap and mat
- Safety glasses
- Heat-resistant gloves for handling hot components
- Dust mask when cleaning equipment

### 6.2 Basic Tools

- Precision screwdriver set (Phillips, flat, Torx, hex)
- Tweezers (ESD-safe)
- Pliers (needle-nose, cutting)
- Cable ties and wire strippers
- Multimeter
- Soldering station (temperature-controlled) with various tips
- Hot air rework station
- Solder (lead-free, various gauges)
- Flux and cleaning solutions
- Desoldering pump or solder wick
- Magnifying lamp or USB microscope
- Thermal paste applicator

### 6.3 Specialized Equipment

- ASIC chip tester (if available)
- Hash board test fixture
- PSU load tester
- Network cable tester
- IR thermometer or thermal imaging camera
- Oscilloscope (for advanced diagnostics)

- Programmers for EEPROM and flash memory
- DC regulated power supply for bench testing
- WhatsMiner test jig (or equivalent)

### 6.4 Software Tools

- Whatsminer configuration software (WhatsMiner Tool or equivalent)
- Firmware files for various Whatsminer models
- Network scanning tools
- Diagnostic software
- Repair tracking/documentation system
- Reference library of schematics and manuals

## 7. Safety and Compliance

### 7.1 Electrical Safety

- Always disconnect miners from power sources before beginning any repair work
- Use proper lockout/tagout procedures when applicable
- Verify with a multimeter that capacitors are discharged before handling components
- Use only properly grounded outlets and tools
- Never work on wet surfaces or with wet hands
- Ensure proper ventilation when soldering

### 7.2 ESD Protection

- Always wear properly grounded ESD wrist strap when handling components
- Use ESD-safe mats on work surfaces
- Store sensitive components in ESD-safe packaging
- Maintain proper humidity levels in repair area (40-60% relative humidity)
- Use ESD-safe tools when working with sensitive components

### 7.3 Thermal Safety

- Allow hot components to cool completely before handling
- Use heat-resistant gloves when necessary
- Maintain proper ventilation when using hot air rework equipment
- Keep flammable materials away from heat sources
- Have appropriate fire extinguisher accessible to work area

### 7.4 Chemical Safety

- Store all chemicals (flux, cleaners, etc.) in properly labeled containers
- Refer to Material Safety Data Sheets (MSDS) for all chemicals used
- Use chemicals in well-ventilated areas

- Wear appropriate PPE (gloves, eye protection) when handling chemicals
- Dispose of chemical waste according to local regulations

## 7.5 Ergonomic Considerations

- Maintain proper posture when working at repair bench
- Use adequate lighting to reduce eye strain
- Take regular breaks during extended repair sessions
- Adjust work surface height to comfortable level
- Use magnification tools when working with small components

# 8. Repair Procedures

## 8.1 Diagnostic Process

### 8.1.1 Initial Assessment

1. Record miner details (model, serial number, and reported issue)
2. Visually inspect for obvious damage (burnt components, physical damage)
3. Check for unusual odors that may indicate burnt components
4. Document any visible abnormalities with photographs
5. Review repair history if available

### 8.1.2 Power-On Diagnostics (When Safe)

1. Connect the miner to power supply (but not to mining pool)
2. Observe startup behavior including:
    - Fan operation
    - LED indicators
    - Unusual sounds
3. Connect to network and access web interface if possible
4. Record error messages, temperatures, and hashrate information
5. Run built-in diagnostic tests if available

### 8.1.3 Component Isolation

1. Determine if issue is with:
    - Power supply
    - Control board
    - Hash boards
    - Fans
    - Firmware
    - Cables and connections
2. Test components individually when possible
3. Swap known good components to isolate faulty ones

### 8.1.4 Documentation of Findings

1. Record all diagnostic steps performed
2. Document identified issues
3. Create repair plan including parts needed
4. Estimate repair time and cost
5. Get approval for repairs if required

## 8.2 Hash Board Repairs

### 8.2.1 Hash Board Testing

1. Visually inspect hash board for damaged components
2. Check for burnt areas, broken traces, or damaged ASIC chips
3. Test voltage at test points using multimeter
4. Connect hash board to test fixture if available
5. Measure resistance at key points to identify shorts

### 8.2.2 ASIC Chip Replacement

1. Identify faulty ASIC chip(s) through testing
2. Apply flux to chip leads
3. Use hot air rework station to heat chip evenly (350-380°C)
4. Remove chip carefully when solder liquefies
5. Clean pad area with solder wick
6. Apply solder paste to pads
7. Position new chip precisely using tweezers
8. Reflow solder using hot air station
9. Clean area with isopropyl alcohol
10. Inspect solder joints with magnification
11. Test repaired board

### 8.2.3 Capacitor and Resistor Replacement

1. Identify faulty component through testing
2. Apply flux to component leads
3. Remove component using soldering iron or hot air
4. Clean pad area
5. Apply small amount of solder to one pad
6. Position new component
7. Solder one lead first, then the second
8. Clean area with isopropyl alcohol
9. Verify correct resistance/continuity with multimeter

### 8.2.4 Trace Repair

1. Identify broken trace using continuity testing

 

2. Clean area thoroughly
3. Scrape away solder mask to expose copper on both sides of break
4. Apply flux
5. Solder a small jumper wire or use conductive pen to bridge gap
6. Clean area with isopropyl alcohol
7. Test continuity after repair
8. Apply conformal coating or nail polish to protect repair

## 8.3 Control Board Repairs

### 8.3.1 Control Board Assessment

1. Check for visible damage (burnt components, physical damage)
2. Test voltage at key test points
3. Verify network port functionality
4. Test communication with hash boards
5. Verify flash memory integrity if possible
6. Test PIC functionality if applicable

### 8.3.2 Network Interface Repair

1. Test Ethernet port functionality
2. Check for damaged transformer or connector
3. Replace Ethernet jack if physically damaged
4. Test network connectivity after repair

### 8.3.3 Flash Memory Replacement/Reprogramming

1. Identify corrupted firmware through diagnostic tests
2. Back up existing firmware if possible
3. Remove flash memory chip if necessary
4. Program replacement chip or reprogram existing chip
5. Install chip and verify proper operation

### 8.3.4 Component-Level Repair

1. Identify faulty components through testing
2. Replace damaged capacitors, resistors, or ICs
3. Check voltage regulator functionality
4. Verify clock signal generation
5. Test repaired board for proper operation

## 8.4 Power Supply Unit Repairs

### 8.4.1 PSU Assessment

1. Inspect for physical damage or burnt components

2. Check input and output connections
3. Test fan operation
4. Measure output voltages under no load
5. Test output voltages under load if equipment available

### 8.4.2 PSU Repair Guidelines

1. CAUTION: PSU repair involves high voltage components. Only attempt if qualified.
2. Discharge capacitors before working on PSU
3. Check fuses and replace if blown
4. Test and replace faulty capacitors
5. Check for cold solder joints and reflow if necessary
6. Test repaired PSU thoroughly before reinstallation

### 8.4.3 PSU Replacement Criteria

1. Replace PSU if:
    - Output voltages are out of specification
    - Internal components show significant damage
    - Repair would cost more than 50% of replacement
    - Unit is older than 2 years with heavy use
2. Verify replacement PSU compatibility with miner model
3. Record old and new PSU serial numbers

## 8.5 Fan Replacement and Repair

### 8.5.1 Fan Assessment

1. Inspect fans for physical damage
2. Test fan operation at various speeds
3. Listen for unusual noises (grinding, clicking)
4. Measure fan RPM if possible
5. Check fan power connections

### 8.5.2 Fan Repair or Replacement

1. Clean fan blades and housing with compressed air
2. Lubricate bearings if accessible
3. Replace fans that:
    - Fail to start reliably
    - Run at inconsistent speeds
    - Make excessive noise
    - Show physical damage
4. Verify replacement fan specifications match original
5. Test fan operation after replacement

 

## 8.6 Firmware Issues

### 8.6.1 Firmware Assessment

1. Check current firmware version
2. Verify compatibility with hardware
3. Review error messages in logs
4. Test basic functionality
5. Check for known issues with current version

### 8.6.2 Firmware Update/Repair

1. Download correct firmware for specific model
2. Verify firmware file integrity (checksum)
3. Back up current configuration settings
4. Follow manufacturer's update procedure
5. Monitor update process closely
6. Verify successful update and proper operation
7. Restore configuration settings if needed

### 8.6.3 Recovery from Failed Update

1. Attempt recovery mode access if available
2. Use emergency recovery procedures if standard update fails
3. Reprogram flash memory directly if necessary
4. Replace control board if firmware cannot be recovered
5. Document recovery process used

# 9. Model-Specific Considerations

## 9.1 M20/M21 Series Specific Procedures

1. Common Issues:
    - Control board communication errors
    - PIC controller issues
    - Temperature sensor failures
2. Repair Techniques:
    - Check crystal oscillator functionality
    - Test and replace temperature sensors
    - Check firmware compatibility with hardware revision

## 9.2 M30/M31 Series Specific Procedures

1. Common Issues:
    - Power delivery issues
    - Hash board communication errors

    - Cooling system optimization
2. Repair Techniques:
    - Test voltage distribution across hash boards
    - Check ribbon cable connections and integrity
    - Optimize thermal paste application
    - Verify correct fan operation sequence

### 9.3 M50 Series Specific Procedures

1. Common Issues:
    - Advanced cooling system failures
    - Specialized power management issues
    - Firmware compatibility
2. Repair Techniques:
    - Test liquid cooling components if applicable
    - Check ASIC power management systems
    - Verify firmware compatibility with hardware revision
    - Test upgraded power delivery systems

## 10. Quality Control and Testing

### 10.1 Functional Testing

1. Power on repaired miner in test environment
2. Verify all fans operate correctly
3. Check network connectivity
4. Access web interface and verify settings
5. Check that all hash boards are detected
6. Monitor temperatures during test operation
7. Verify hashrate meets specifications

### 10.2 Stress Testing

1. Run miner at full capacity for minimum 2 hours
2. Monitor temperatures throughout test
3. Check for stable hashrate
4. Verify no error messages appear
5. Monitor power consumption
6. Check for unusual sounds or vibrations

### 10.3 Quality Assurance Checklist

1. All screws and fasteners properly installed and tight
2. All cables properly connected and secured
3. No loose or missing components

4. No visible damage to exterior
5. All labels and serial numbers intact
6. Firmware updated to recommended version
7. Test results documented
8. Repair actions documented

### 10.4 Final Inspection

1. Clean exterior of dust and fingerprints
2. Verify all access panels are properly secured
3. Apply a quality control sticker with date and technician ID
4. Package in appropriate anti-static packaging
5. Complete final documentation
6. Transfer to shipping/storage or return to customer

## 11. Parts Inventory Management

### 11.1 Parts Tracking

1. Maintain database of all parts used in repairs
2. Record part numbers, quantities, and costs
3. Track usage rates to identify common failures
4. Document compatibility between different models
5. Record supplier information for each part

### 11.2 Inventory Control

1. Establish minimum stock levels for common parts
2. Conduct weekly inventory counts
3. Place orders when inventory reaches reorder point
4. Rotate stock to use oldest components first
5. Store components in appropriate anti-static packaging
6. Maintain climate-controlled storage for sensitive components

### 11.3 Sourcing and Procurement

1. Validate authenticity of parts from suppliers
2. Maintain list of approved suppliers
3. Document specifications for acceptable substitute parts
4. Evaluate cost vs. quality for alternate suppliers
5. Track lead times for critical components

### 11.4 Defective Parts Handling

1. Tag defective parts removed from miners

2. Record serial numbers and failure modes
3. Store RMA-eligible parts separately
4. Process RMA returns according to manufacturer guidelines
5. Properly dispose of non-returnable parts

# 12. Documentation and Record-Keeping

## 12.1 Repair Documentation

1. Create repair ticket for each miner
2. Record miner details:
    - Model and serial number
    - Firmware version
    - Reported issue
3. Document diagnostic findings
4. List all components replaced
5. Record test results before and after repair
6. Note any remaining issues or recommendations

## 12.2 Repair History Tracking

1. Maintain database of all repairs by serial number
2. Track recurring issues by miner and batch
3. Calculate repair success rates
4. Monitor longevity of repairs
5. Identify patterns in failures

## 12.3 Reporting

1. Generate daily repair completion reports
2. Track repair turnaround time
3. Report on parts usage and inventory levels
4. Document repair success rates
5. Highlight recurring issues that may indicate batch problems

## 12.4 Knowledge Base Maintenance

1. Document new repair techniques discovered
2. Create and update troubleshooting guides
3. Maintain library of known issues by model
4. Document workarounds for common problems
5. Share knowledge with repair team

# 13. Environmental Considerations

## 13.1 Work Environment

1. Maintain clean, dust-free repair area
2. Control temperature (20-25°C) and humidity (40-60%)
3. Provide adequate lighting and magnification
4. Install proper ventilation for soldering and rework
5. Use anti-fatigue mats at workstations

## 13.2 Waste Management

1. Separate electronic waste for proper recycling
2. Dispose of chemicals according to local regulations
3. Recycle packaging materials when possible
4. Maintain documentation of waste disposal
5. Follow environmental best practices

## 13.3 Energy Efficiency

1. Use energy-efficient test equipment
2. Power down test systems when not in use
3. Optimize test procedures to minimize power consumption
4. Track energy usage in repair operations
5. Identify opportunities for reducing energy consumption

# 14. Repair Benchmarks and Performance Standards

## 14.1 Time Standards

1. Initial diagnosis: 30-60 minutes
2. Hash board repair: 1-3 hours depending on complexity
3. Control board repair: 1-2 hours
4. PSU repair or replacement: 30-60 minutes
5. Fan replacement: 15-30 minutes
6. Firmware update/repair: 30-60 minutes
7. Final testing: minimum 2 hours

## 14.2 Quality Standards

1. Repaired miners must achieve minimum 95% of rated hashrate
2. Temperature under load must not exceed manufacturer specifications
3. All functions must operate correctly
4. Repair must remain effective for minimum 30 days
5. Cosmetic condition should match or exceed pre-repair condition

### 14.3 Technician Performance Metrics

1. Repair success rate (minimum 90%)
2. Average time per repair type
3. Parts usage efficiency
4. Documentation accuracy
5. Return/callback rate (should be under 5%)
6. Knowledge sharing contribution

## 15. Training and Skill Development

### 15.1 Initial Training

1. Safety procedures and PPE use
2. ESD protection protocols
3. Diagnostic methods and testing
4. Soldering and rework techniques
5. Documentation and record-keeping
6. Model-specific repair procedures

### 15.2 Ongoing Training

1. New model repair procedures
2. Advanced troubleshooting techniques
3. New tool and equipment operation
4. Firmware update procedures
5. Quality improvement methods

### 15.3 Skill Assessment

1. Regular evaluation of repair quality
2. Technical knowledge testing
3. Hands-on skill demonstration
4. Documentation review
5. Peer review of repair techniques

### 15.4 Cross-Training

1. Ensure multiple technicians can perform each repair type
2. Rotate responsibilities to build broad knowledge
3. Share specialized skills among team members
4. Document unique techniques for knowledge transfer
5. Mentor new technicians

# 16. Appendices

## 16.1 Model-Specific Reference Materials

1. Pinout diagrams for common Whatsminer models
2. Voltage reference charts
3. Component location guides
4. Common failure points by model
5. Test point guides

## 16.2 Troubleshooting Decision Trees

1. No power symptoms
2. Low hashrate issues
3. Overheating problems
4. Network connectivity failures
5. Fan failure symptoms

## 16.3 Parts Cross-Reference

1. Compatible replacement components by model
2. Alternative part numbers
3. Supplier information
4. Lead time expectations
5. Price comparisons

## 16.4 Forms and Checklists

1. Intake inspection form
2. Repair documentation template
3. Quality control checklist
4. Final inspection checklist
5. RMA processing form

# 17. References

## 17.1 Manufacturer Documentation

1. MicroBT technical manuals
2. Service bulletins
3. Firmware update guides
4. Specifications and datasheets
5. Warranty information

 

## 17.2 Technical References

1. Electronic component datasheets
2. Soldering and rework best practices
3. ESD protection standards
4. Testing procedures and standards
5. Industry best practices

## 17.3 Safety Standards

1. Electrical safety guidelines
2. Chemical handling procedures
3. Fire safety protocols
4. Ergonomic standards
5. Personal protective equipment specifications

APPROVED BY: ___________________________ DATE: ___________

EFFECTIVE DATE: ___________

REVIEW DATE: ___________

DOCUMENT VERSION: 1.0
Note: This SOP should be reviewed and updated annually or whenever significant changes to equipment, procedures, or safety requirements occur.

 